



Deceptive Reviews Of Indian Educational Systems

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Project Overview

- ▶ Objective: Detect fake/genuine educational reviews using Machine Learning and NLP.
- ▶ Domain: Indian education review platforms.
- ▶ Platforms: Career360 & College dunia.

Background of the Project

- ▶ Online reviews influence student decisions.
- ▶ Fake reviews can mislead students about institutions.
- ▶ Need for automated fake review detection.

Problem Statement



- ▶ Fake reviews distort college perception.
- ▶ Students may choose wrong institutions.
- ▶ Need affordable automated review verification.

Objectives of the System

- ▶ Collect education reviews
- ▶ Preprocess text
- ▶ TF-IDF + Logistic Regression classification
- ▶ VADER sentiment analysis
- ▶ Dashboard visualization

Scope of the Project

- ▶ Data scope: 855 reviews
- ▶ Technical scope: Python, Scikit-learn, NLTK, Power BI
- ▶ Exclusions: No real-time API deployment

Existing Systems

- ▶ Rule-based filtering
- ▶ Machine Learning: SVM, Naive Bayes, Random Forest
- ▶ Deep Learning: LSTM, BERT
- ▶ Platform-native systems

Proposed System

- ▶ Pipeline:
- ▶ Step 1: Data Collection
- ▶ Step 2: Preprocessing
- ▶ Step 3: TF-IDF
- ▶ Step 4: Logistic Regression
- ▶ Step 5 : Sentiment Analysis
- ▶ Step 6 : Power BI Dashboard

Technologies Used

- ▶ Python
- ▶ Pandas
- ▶ Scikit-learn
- ▶ NLTK + VADER
- ▶ Jupyter Notebook
- ▶ Power BI

Literature Review



- ▶ Ott et al. (2011)
- ▶ Li et al. (2014)
- ▶ Jindal & Liu (2008)
- ▶ Mukherjee et al. (2012)
- ▶ Indian education review studies

Comparative Analysis

- ▶ Compare Logistic Regression, SVM, Random Forest, LSTM, BERT.
- ▶ Selected model: Logistic Regression + TF-IDF due to efficiency and interpretability.

SDLC Model

- ▶ Agile Model selected for iterative ML experimentation.
- ▶ Also reviewed Waterfall, Prototype and Spiral models.

Functional Requirements

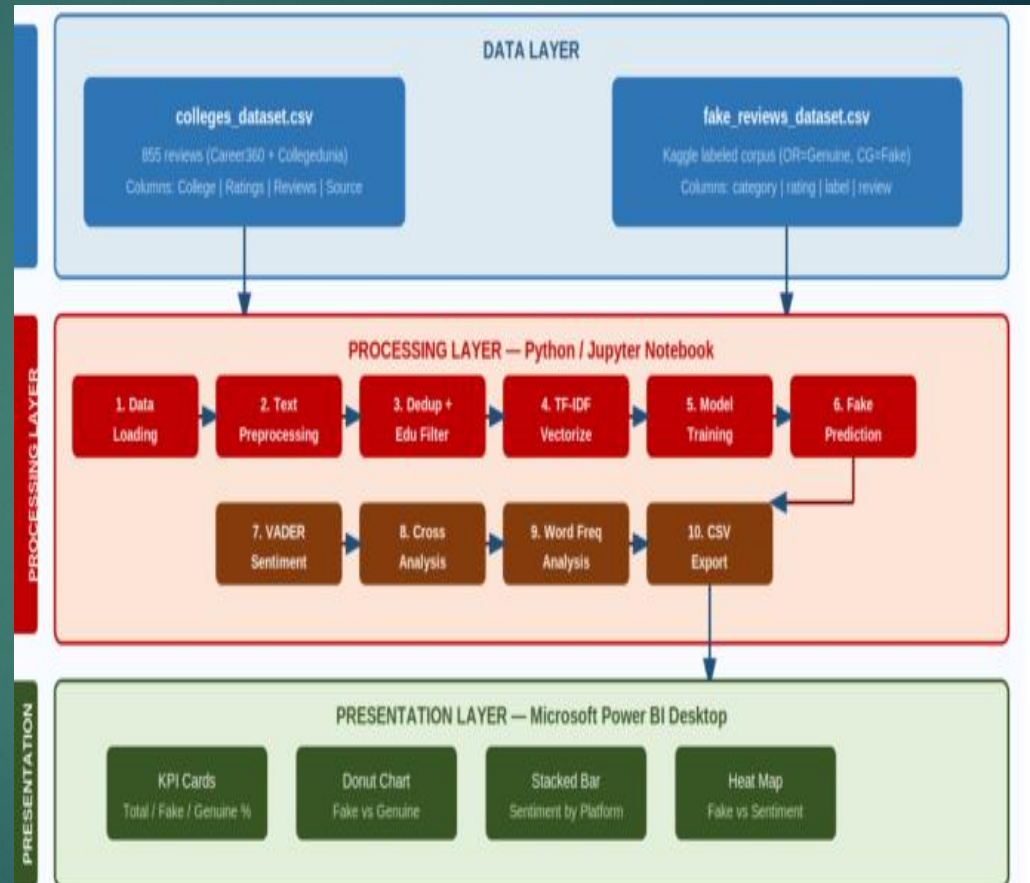
- ▶ Data loading
- ▶ Preprocessing
- ▶ Model training
- ▶ Prediction
- ▶ Sentiment scoring
- ▶ Export to CSV

Non-Functional Requirements

- ▶ Performance
- ▶ Accuracy
- ▶ Reproducibility
- ▶ Scalability
- ▶ Usability
- ▶ Security

System Architecture

- ▶ Three Layers:
- ▶ 1. Data Layer
- ▶ 2. Processing Layer
- ▶ 3. Presentation Layer



Data Preprocessing Pipeline

- ▶ Lowercase conversion
- ▶ URL & HTML removal
- ▶ Tokenization
- ▶ Stop word removal
- ▶ Text normalization

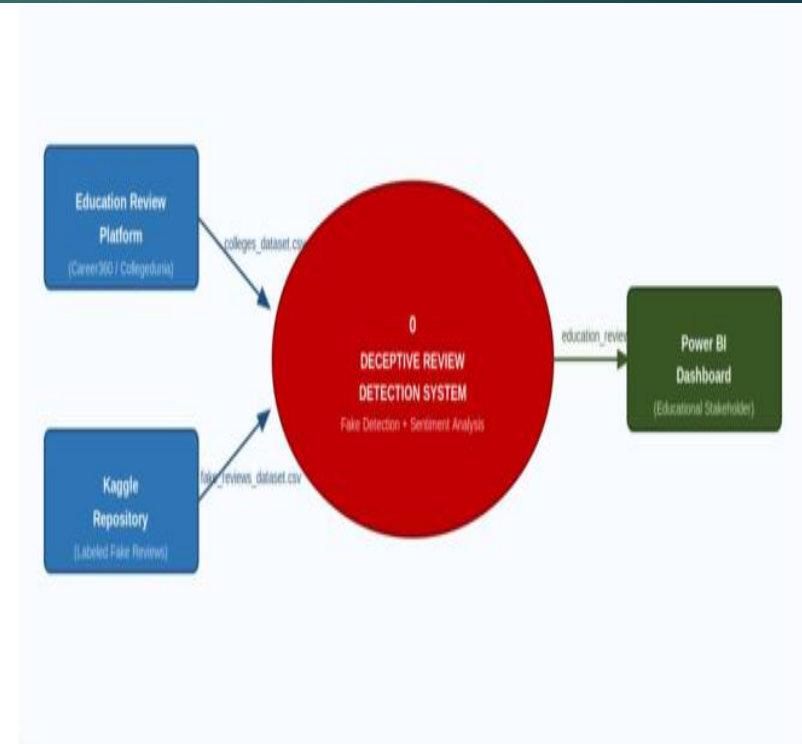


Figure 3.2: Level 0 Data Flow Diagram (Context Diagram)

Machine Learning Model

- ▶ TF-IDF Vectorization (5000 features)
- ▶ Logistic Regression Classifier
- ▶ 80/20 Train-Test Split

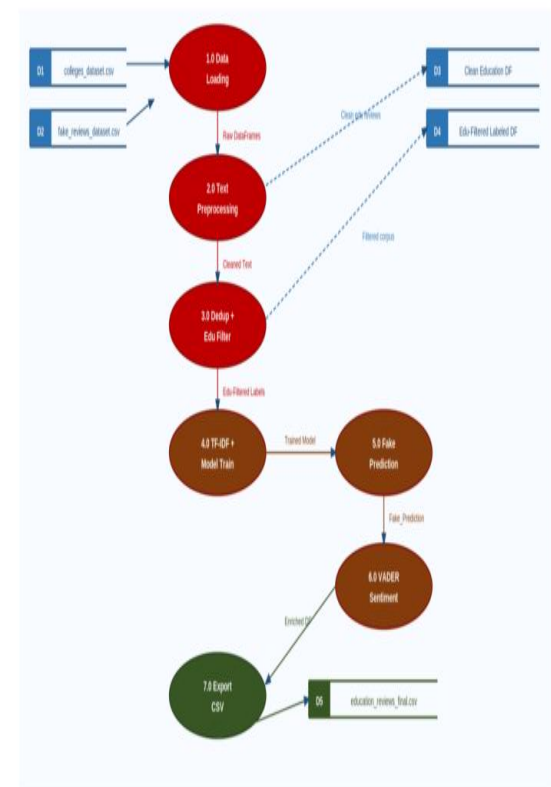
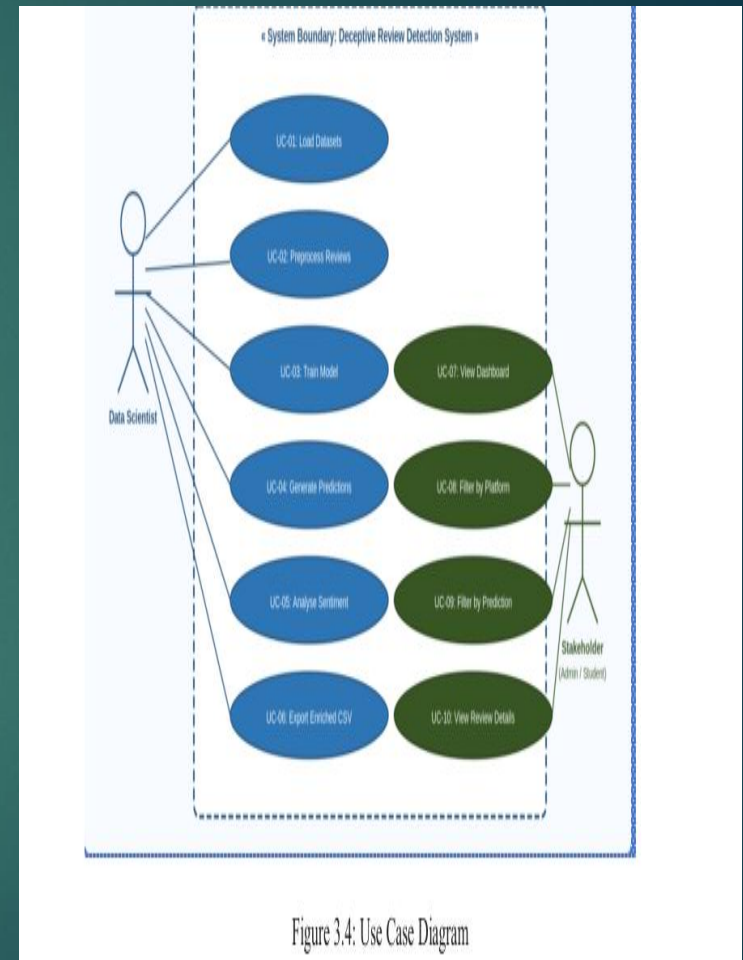


Figure 3.3: Level 1 Data Flow Diagram

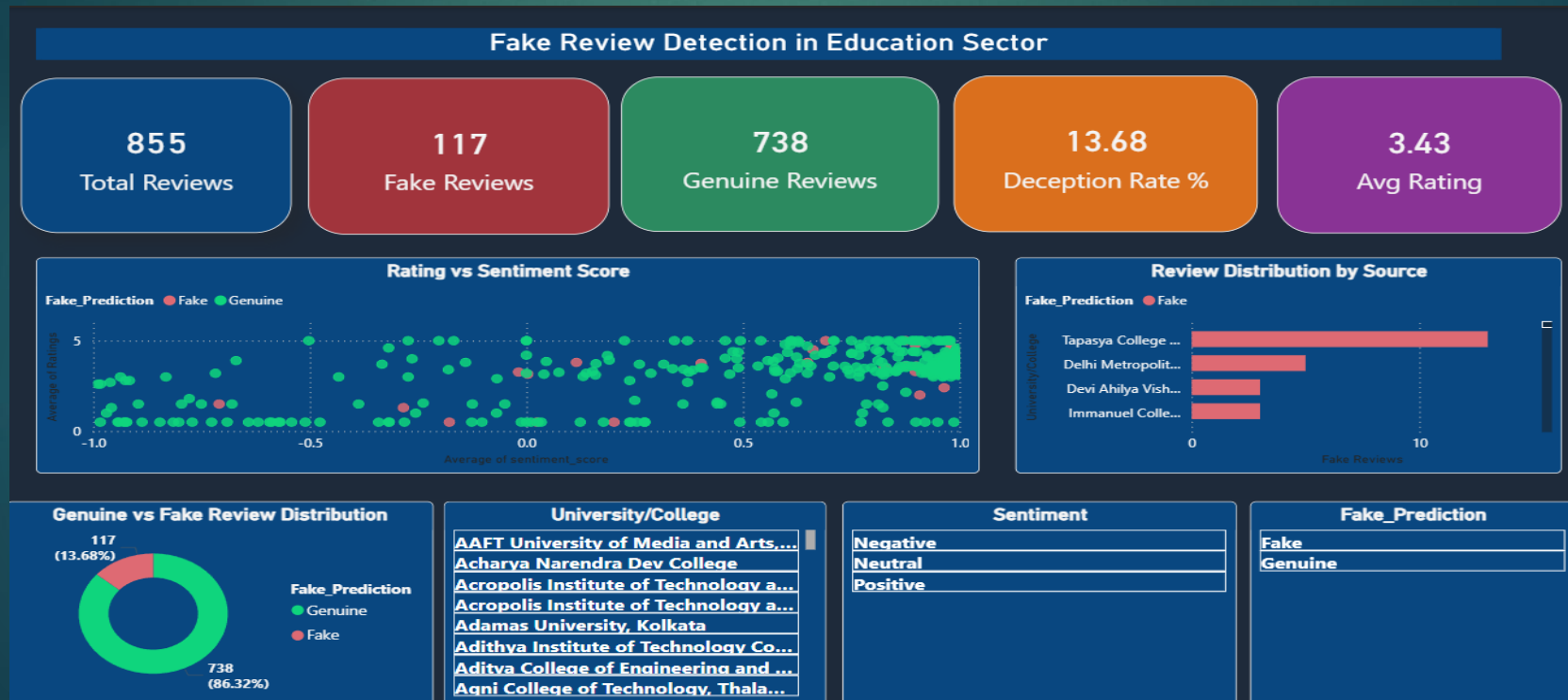
Sentiment Analysis

- ▶ VADER Sentiment Analyzer
- ▶ Positive (>0.05)
- ▶ Neutral (-0.05 to 0.05)
- ▶ Negative (<-0.05)



Results & Dashboard

- ▶ Fake vs Genuine classification
- ▶ Sentiment distribution
- ▶ Cross-analysis in Power BI dashboard



Conclusion & Future Scope

- ▶ Project improves trust in educational reviews.
- ▶ Future: Real-time API, BERT integration, live scraping, larger datasets.